

# Chapter 4

# Environmental Management



**1. Reorganize the following food chain. Describe the ecosystem to which it belongs.**

**Grasshopper – Snake – Paddy field – Eagle – Frog**

**Paddy field — Grasshopper — Frog — Snake — Eagle**

- i. Paddy field represents the primary producer which are plants.
- ii. Grasshopper is a primary consumer. Primary consumers are organisms that eat the primary producers and are usually herbivores.
- iii. Frog is a secondary consumer. Secondary consumers are organisms that eat the primary consumers and are generally meat-eaters/carnivores.
- iv. Snake is a tertiary consumer. Tertiary consumers are organisms that eat the secondary consumers. These organisms are carnivore-eating carnivores, like snakes or big fish.
- v. Eagle is a quaternary consumer, and, in this case, it is the apex consumer because it occupies the highest position in the food chain.

**2. Explain the statement- 'we have got this Earth planet on lease from our future generations and not as an ancestral property from our ancestors.'**

- i. Existence of man is entirely dependent on the existence of ecosystem.
- ii. Man derives his benefits by using various components of the ecosystem.
- iii. We can get these materials easily from the nature as our ancestors protected and conserved them for us.

- iv. Today man has proved his superiority as compared to other animals. He has indiscriminately used the natural resources for personal greed.
- v. Due to this, future generations may not benefit from the nature. They will suffer from pollution, drought, unseasonal rains, green house effect etc.
- vi. Thus, it is our responsibility to preserve, conserve and protect nature so that next generations will get conducive climate and their life may not be in trouble.
- vii. Thus, it is rightly said that “we got the planet earth on lease from our future generations and not ancestral property”.
- viii. Nature is not our ancestral property, but we have to take care of it so that it can be handed over to the next generations without any further damage to it.



### 3. Write short notes.

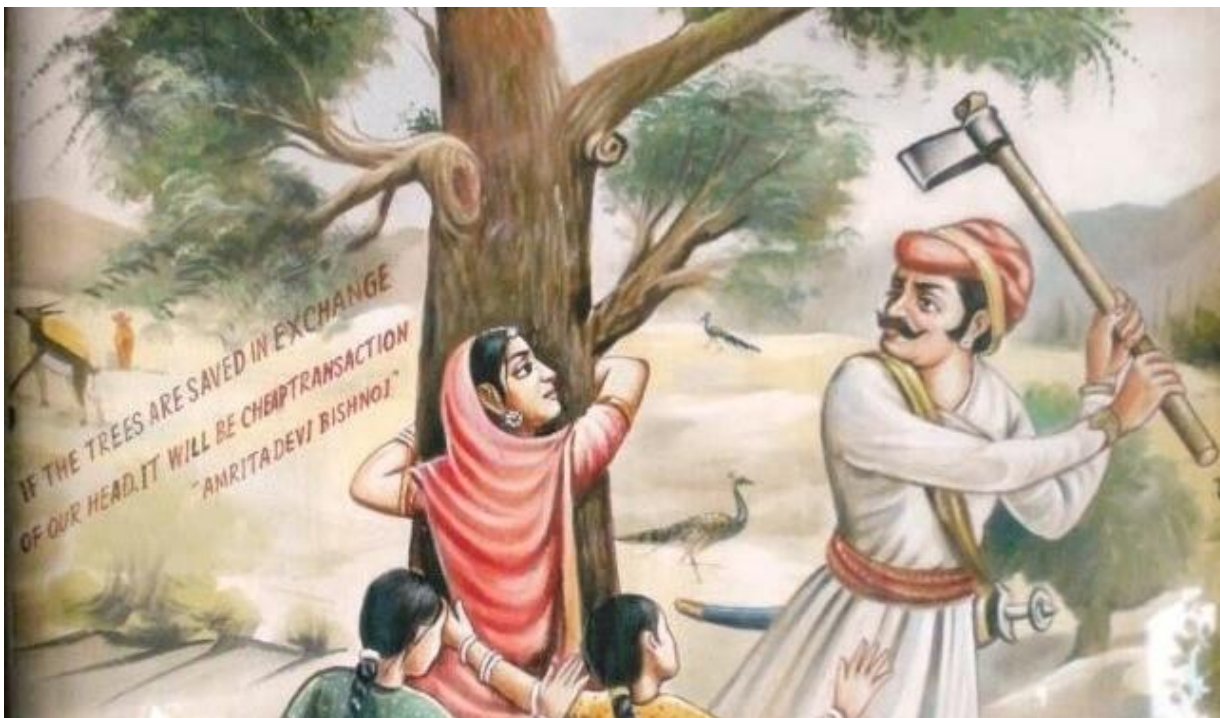
#### a. Environmental Conservation.

- i. Since existence of human, there is interrelationship between human and environment.
- ii. Human stepped on the Earth long after its formation and proved its superiority as compared to other animals with the help of characters like intelligence, memory, imaginary ability, etc.
- iii. Human established domination over the nature and utilized all the natural resources as much as possible.
- iv. To live a satisfactory life, human kept snatching from nature as much as possible and this led to increase in problems.
- v. If human has disturbed the environmental balance, then human itself only can conserve and improve the quality of nature.
- vi. It will be possible to answer the environmental problems only if environmental protection-conservation becomes an effective public movement.
- vii. For this purpose, values like positive attitude and affection towards environment, knowledge about it, etc. should be inculcated among the children since their childhood.
- viii. All the developed, developing and underdeveloped countries have accepted the responsibility of environmental conservation.
- ix. They have defined the future plans about environmental protection and have constituted the necessary laws.



## b. Chipko Movement of Bishnoi.

- i. In the 1970s, an opposition to the destruction of forests spread throughout India and came to be known as the Chipko movement.
- ii. The name of the movement comes from the word 'embrace' as the villagers hugged the trees and prevented the contractors from cutting them.
- iii. Many communities in India have helped save nature. One such is the Bishnoi community of Rajasthan.
- iv. The original Chipko movement was started about 260 years back in Rajasthan by this community. A large group of villagers led by a lady called Amrita Devi laid down their lives in an effort to protect the Khejri trees from being felled on the orders of the Maharaja of Jodhpur.
- v. After realizing the mistake, the king ordered stoppage of the felling of trees. Honouring the courage of the Bishnoi community, the ruler of Jodhpur, Maharaja Abhay Singh, apologized.
- vi. He issued a royal decree to protect trees and wildlife.



### c. Biodiversity

Biodiversity is the richness of living organisms in nature due to presence of varieties of organisms, ecosystems and genetic variations within a species. Biodiversity occurs at three different levels.

- i. Genetic Diversity : Occurrence of diversity among the organisms of same species is genetic diversity. For example, each human being is different from the other.
- ii. Species Diversity : Innumerable species of organisms occur in nature. This is called as species diversity. Species diversity includes various types of plants, animals and microbes.
- iii. Ecosystem Diversity : Many ecosystems are present in each region. Ecosystem is formed through the interaction between plants, animals, their habitat and changes in the environment. Each ecosystem has its own characteristic animals, plants, microbes and abiotic factors.



#### **d. Sacred groves**

- i. The forest conserved in the name of god and considered to be sacred is called as sacred grove.
- ii. These are in fact 'sanctuaries' conserved by the society and not by the government forest department.
- iii. As it has been conserved in the name of god, it has special protection.
- iv. These clusters of thick forests are present not only in Western Ghats of India but in the entire country.
- v. More than 13000 sacred groves have been reported in India.



#### **e. Disaster and its management.**

- i. Various dangerous events occur many times in the environment. These are called disasters.
- ii. These are 'all of a sudden' troubles to the mankind. Such events cause sudden changes in the environment and thereby, cause damage to it.
- iii. Disaster is a sudden event that leads to huge loss of life and property.
- iv. Disaster may be minor or major, short-time or long-time, it should be overcome and effective disaster management is necessary for it.



- v. Disaster management is either prevention of disasters or making arrangements to face them or at least achieve the abilities to face them.
- vi. Disasters are never planned but losses due to them can be prevented in a planned manner.
- vii. Disaster management is achieving or time to time improving the ability to face the disaster through scientific and careful observations and analysis of data.
- viii. For example, preparing the action plan through study of various aspects like preventive measures, rehabilitation and reconstruction and executing that plan is management of disaster.
- ix. Increasing awareness about disaster management among the general public through training programs, mass media etc. will enable them to be always prepared for disaster rescue.

#### **4. How will you justify that overcoming the pollution is a powerful way of environmental management?**

- i. Environmental balance is maintained through continuous operation of various natural cycles like biogeochemical cycles.
- ii. When some natural factors of environment and some artificial polluted factors damage the environment, it creates imbalance between various factors of the environment and ultimately affects the existence of biotic factors.



- iii. We know that the major cause of environmental damage is pollution.
- iv. Pollution of air, water bodies, soil etc. all contribute to various environmental problems. The continuous increase in pollution has made the environment nearly unfit for us.
- v. This can largely be attributed to the rapid rate at which trees are being cut. In the absence of trees, the level of carbon dioxide in the atmosphere increases, adding to the problem of global warming.
- vi. Moreover, the emission of carbon monoxide, hydrocarbons and other such gases, from vehicles and factories, is rapidly polluting the air making it even difficult to breathe.
- vii. Pollution of water bodies have resulted in the death of aquatic life and making most of the water bodies unfit for human and animal use.
- viii. All the above points highlight the fact that pollution is one of the causes of environmental degradation.
- ix. Environmental management can help solve this issue by efficiently managing the problems associated with pollution.
- x. We need to remember that nature is a treasure which needs to be preserved for our own survival.
- xi. If we do not pay attention to this problem and make efforts to preserve our environment, this wonderful earth would soon cease to exist

## 5. Which projects will you run in relation to environmental conservation? How?

Environmental conservation is a broad term which refers to the actions we take in order to protect our planet and conserve its natural resources. There are various projects which can be carried out to conserve our environment:

- i. **Community Clean-Up Campaigns**: Students can organize and participate in community clean-up events to collect litter from parks, beaches, and public spaces. These events raise awareness about the importance of proper waste disposal and help keep the environment clean.
- ii. **Reforestation and Tree Planting**: Reforestation projects involve planting trees in deforested or degraded areas to restore ecosystems and combat climate change. Students can collaborate with environmental organizations, local governments, and schools to organize tree-planting campaigns.
- iii. **Plastic Reduction and Awareness Campaigns**: Students can run campaigns to raise awareness about the harmful effects of single-use plastics on the environment and promote alternatives. They can conduct workshops, create educational materials, and work with local businesses to reduce plastic usage.

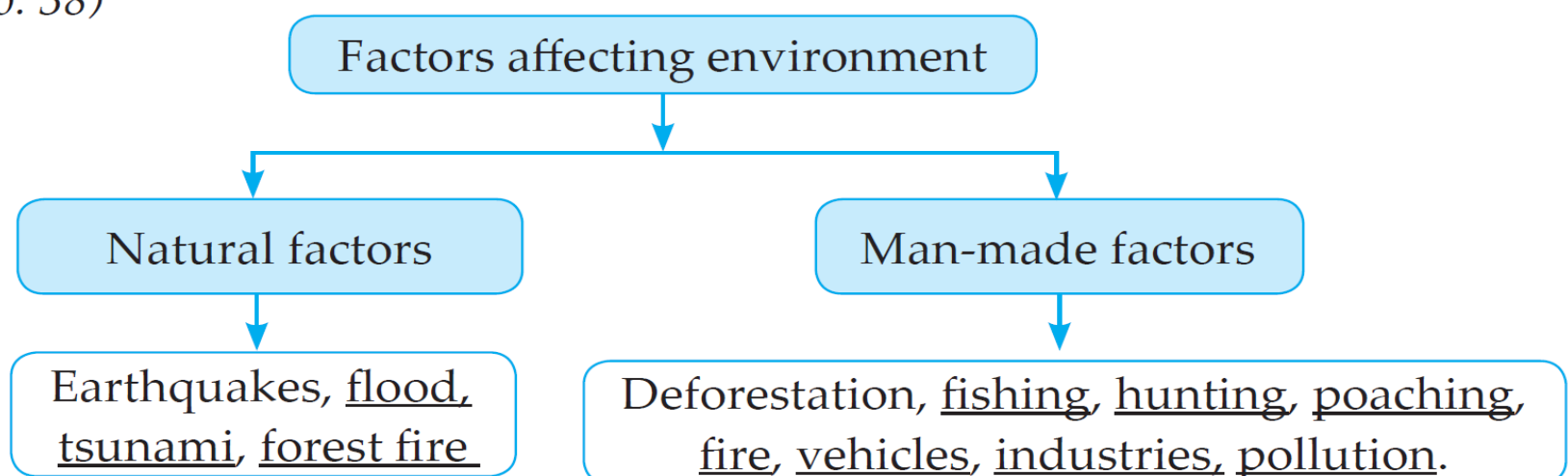
iv. **Save water**: In this project, people and students will be made aware of the importance of water. At the same time, various methods of water conservation will be taught.

## 6. Answer the following.

### a. Write the factors affecting environment.

- i. The biotic and abiotic- factors affect the environment. Among abiotic factors, the physical and chemical factors can alter the conditions of the environment.
- ii. Abiotic factors are either natural or man-made.
- iii. The various interrelationships between different living organisms can also affect the environment.
- iv. The natural factors such as earthquakes, floods, cyclones, cloud bursting, tsunami, forest fires etc. change the environment.
- v. The man-made factors such as deforestation, fishing, hunting, poaching, fire, vehicles, industries, etc. cause permanent damage to the ecosystems. Due to human impact, there can be large-scale changes in the environment.

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## b. Why do human beings have an important place in environment?

- i. Since the existence of human, there is an interrelationship between human and environment.
- ii. Human stepped on the Earth long after the formation of Earth. On the Earth, human being proved its superiority as compared to other animals with the help of characters like intelligence, memory, imaginary ability, etc.
- iii. Human established domination over the nature. Human utilized all the natural resources as much as possible.
- iv. In an attempt to live a satisfactory life, human kept snatching from nature as much as possible and this led to an increase in problems.
- v. If human has disturbed the environmental balance, then human itself only can conserve and improve the quality of nature.
- vi. From this entire scenario, we can understand that human beings have an important place in environment.





### c. Write the types and examples of biodiversity.

- i. Biodiversity is the richness of living organisms in nature due to presence of varieties of organisms, ecosystems and genetic variations within species.
- ii. Biodiversity occurs at three different levels.
  - a) **Genetic Diversity**: Occurrence of diversity among organisms of same species whose members involve in sexual reproduction is genetic diversity.
  - b) **Species Diversity**: Innumerable species of organisms occur in the nature. This is called species diversity. Species diversity includes various types of plants, animals and microbes.
  - c) **Ecosystem Diversity**: Many ecosystems are present in each region. Ecosystem is formed through the interactions between plants, animals, microorganisms, and abiotic factors.

### d. How the biodiversity can be conserved?

**Biodiversity can be conserved in the following ways:**

- 1) Protecting rare species of organisms.
- 2) Establishing national parks and sanctuaries.
- 3) Declaring some regions as 'bioreserves'.
- 4) Projects for conservation of special species.
- 5) Conserving all plants and animals.
- 6) Maintaining records of the traditional knowledge.
- 7) Obeying the rules.

## e. What do we learn from the story of Jadav Molai Payeng?

- i. Jadav Molai Payeng is a highly capable person in Assam. Born in 1963, he is working as a forest worker since the age of 16 years.
- ii. In 1979, the local Social Forestry Department began a social afforestation project on 200 hectares of land.
- iii. 'Molai' was one of the few forest workers who were looking after that project.
- iv. Molai continued to plant the trees even after completion of the project.
- v. As a result of his continuous work of planting and caring for the trees, the barren area witnessed the forest cover over the 1360 acres.
- vi. From his story, it is clear that many people can come together to destroy a forest, but a single person, if determined, can establish a new forest!
- vii. Also, if one person can make such an impact, many people coming together in unity to take efforts for environment conservation can bring about a revolution.



**f. Write the names of biodiversity hot spots.**

- i. A biodiversity hotspot is a biogeographic region with significant levels of biodiversity that is threatened with destruction.
- ii. 34 highly sensitive biodiversity hotspots are reported all over the world.
- iii. Such areas had once occupied 15.7% area of the Earth.
- iv. At present, 86% of the sensitive areas are already destroyed.
- v. Only 2.3% area of the Earth has been left over with sensitive spots. It includes 1,50,000 plant species which are 50% of the world count.
- vi. As far as India is considered, out of 135 species of animals, 85 species are found in the jungles of eastern region.
- vii. About 1,500 endemic plant species are found in western ghat.
- viii. The biodiversity hotspots in India are as follows:
  - 1. The Western Ghats.
  - 2. The Eastern Himalayas.
  - 3. Indo-Burma.
  - 4. Sundaland.



**g. Which are the reasons for endangering the many species of plants and animals? How can we save those?**

Numerous species of animals and plants have become endangered due to activities of human beings.

- i. Animal skin, flesh and bones are used for various purposes. Hence, poaching takes place which becomes a big threat to the existence of animal and plant life.
- ii. Wood, medicinal plants, gums, resins are the plant products which are useful, hence plants are brutally cut.
- iii. Also, animal-human conflicts occur due to invasion of human settlements into the habitats of wild animals.
- iv. Construction of dams, roads, and colonies destroy the habitats of wildlife.
- v. Industrialization, urbanization and population explosion of humans are putting severe pressure on all the existing biodiversity.
- vi. We can save the remaining animal and plant life through environmental education, awareness programs, strict rules and regulations, implementing various environmental acts, and prohibiting poaching can prevent the loss of varieties of plants and animals.
- vii. A stretch of land is protected by declaring it as the sanctuary or a national park by the Government. Even the locals can protect it as a sacred grove.



## 7. What are the meanings of following symbols? Write your role accordingly?

a.



This sign is used for indicating that 'this object can be **reused** and **recycled**'.

Our role in recycling is pivotal for a sustainable future. By segregating materials, reducing waste, and reusing items, we conserve resources and minimize environmental impact. Active participation in recycling programs empowers us to contribute directly to a greener and cleaner planet.

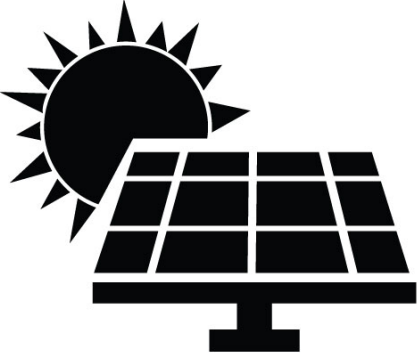
b.



This sign is used to indicate '**Save Water**'.

Our role in saving water is crucial for conservation. By fixing leaks promptly, using water-efficient appliances, and practicing mindful water usage, we help ensure a sustainable water supply for both current and future generations. Small actions collectively make a significant difference in preserving this precious resource.

c. .



This symbol advocates the use of **solar energy**.

Our role in using solar energy is vital for a cleaner future. By installing solar panels, promoting solar initiatives, and reducing dependency on fossil fuels, we contribute to a sustainable energy transition and lessen the effects of climate change. Embracing solar power empowers us to be active participants in the shift towards renewable energy sources.

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# INTEXT QUESTIONS

## 1. What is ecosystem? Which are its different components?

- i. The biotic and abiotic factors and their interactions is called an ecosystem.
- ii. Biotic factors: Plants, animals, microbes  
Abiotic Factors: Water, air, soil, etc.

## 2. Which are the types of consumers? What are the criteria for their classification?

- i. Consumers are classified as primary consumers, secondary consumers and top-level consumers, etc.
- ii. These types are according to the trophic level to which they belong.

## 3. What may be the relationship between lake and birds on tree?



- i. Tree near lake comes under lake ecosystem.
- ii. Birds reside on the trees and may trap prey from pond.

4. What is difference between food chain and food web?

| Food Chain                                                                                                  | Food Web                                                                                                   |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Interactions between producers, consumers and decomposers in a definite sequence is called as a food chain. | The interconnection among different food chains in an ecosystem at various levels is called as a food web. |
| It consists of only one straight chain.                                                                     | It consists of many interconnected food chains.                                                            |

5. Observe the picture and answer the following questions.

- a. Write the name and category of each of the component shown in picture.

Grass (producer), Grasshopper (primary consumer), Frog (Secondary consumer), Snake (Tertiary consumer), Eagle (Apex consumer), Microbes (Decomposers).



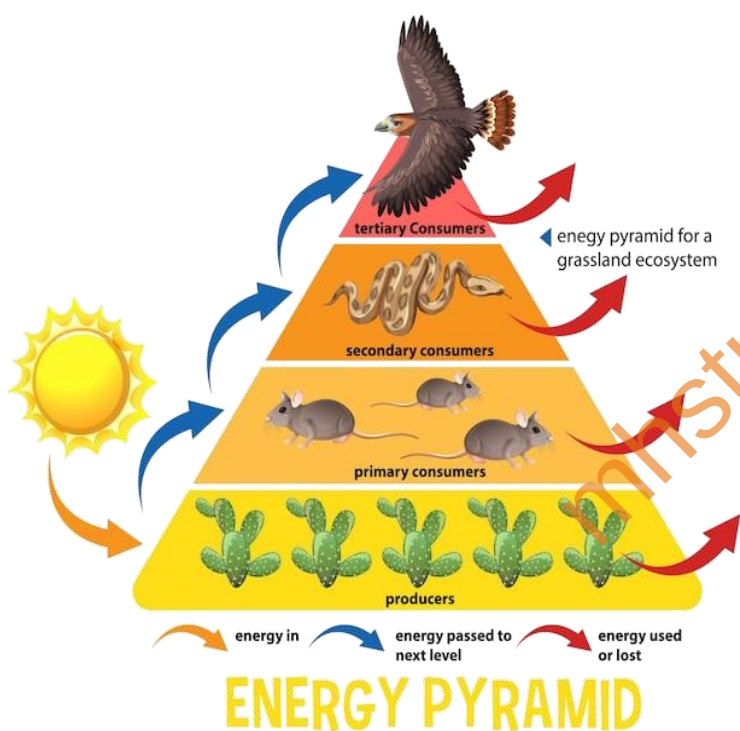
**b. What is necessary to convert this picture into food web? Why?**

- i. Include more primary consumers like squirrel, deer, goat etc.
- ii. Include more secondary consumers like wolf, fox etc.
- iii. Include more tertiary and apex consumers like lion, tiger etc.
- iv. Show interrelationship of these animals to convert the food chains into a food web.

**6. Which are different trophic levels in food chain?**

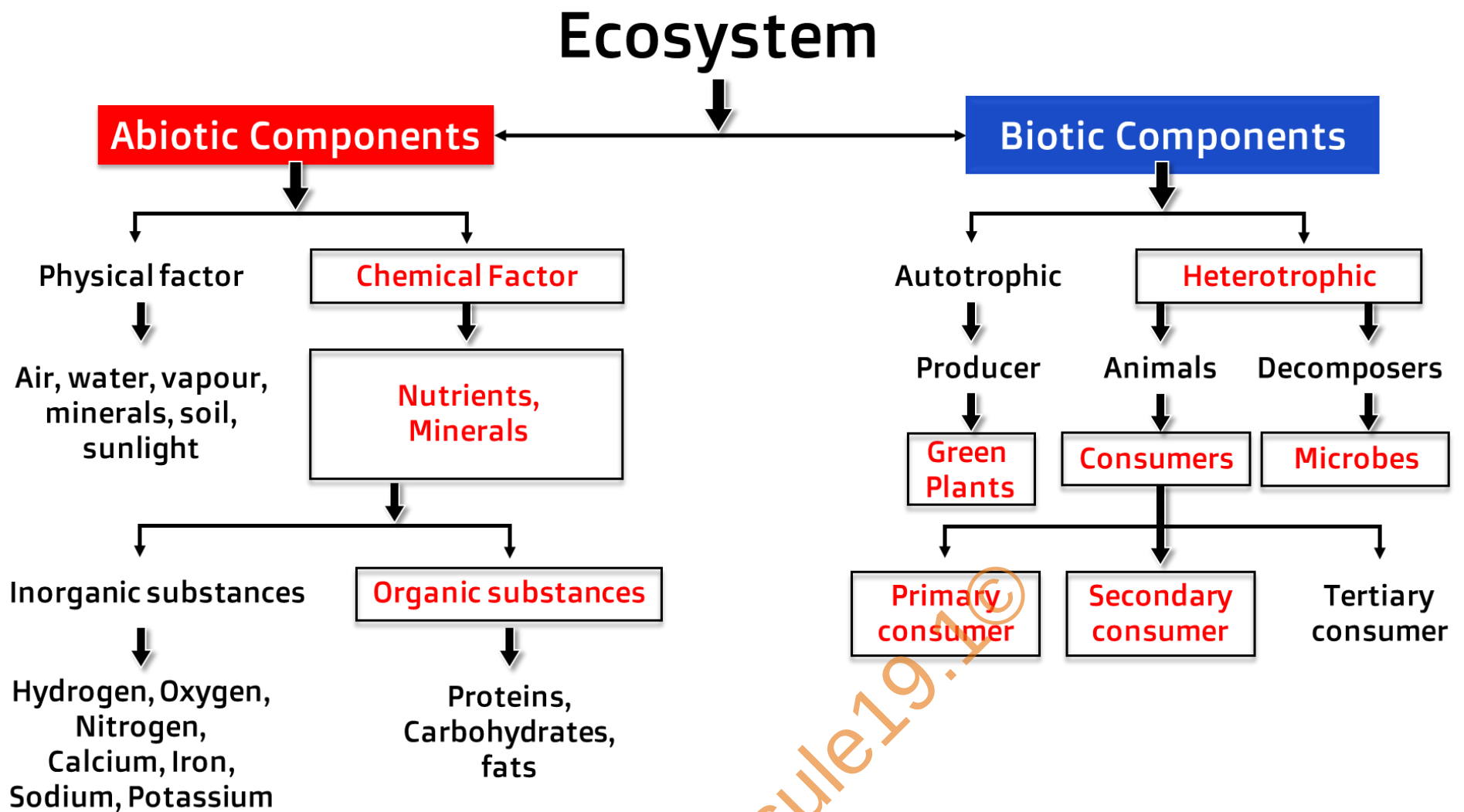
Producers, primary consumers, secondary consumers, tertiary consumers, top level (apex) consumers and decomposers.

**7. What is energy pyramid?**



- i. Energy pyramid is a graphical model which represents the flow of energy in an ecosystem.
- ii. It consists of different levels which represent different trophic levels of food chain.
- iii. The energy pyramid shows the process of energy travelling up a food chain.

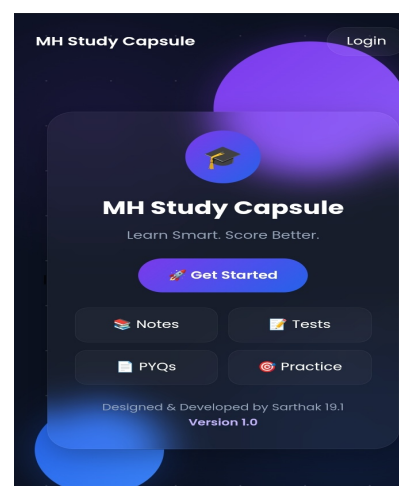
8. Fill up the blank boxes and display the completed chart in classroom.



9. A bird building nest on a tree feeds upon the fishes in nearby pond.

Whether this bird is part of both i.e., tree as well as pond ecosystem?

- Yes, the bird is part of both the tree and pond ecosystems.
- As it builds its nest in the tree and relies on the tree for shelter and support, it is directly linked to the tree ecosystem. It might feed on fruits of the tree as well as insects dwelling there.
- Additionally, by feeding on fishes in the nearby pond, the bird is connected to the pond ecosystem as well.



## **10. If frog population in paddy field declines all of a sudden,**

### **a. What will be the effect on paddy crop?**

Paddy crop production will go down as the population of grasshoppers will increase and they will damage the paddy crop.

### **b. Number of which consumers will decline, and which will increase?**

Primary consumers (Grasshopper) will increase, and tertiary consumers (snake) will decline.

### **c. What will be the overall effect on the ecosystem?**

The natural balance of the ecosystem will be disturbed.

## **11. What is environment?**

- i. Physical, chemical and biological factors affecting the living organisms in any possible way is collectively called as environment.
- ii. In short, environment is the condition in surrounding.

## **12. What is included in environment?**

Environment includes many biotic, abiotic, natural and artificial factors.

## **13. Which cycles are operated in environment? What is their importance?**

- i. Various cycles operated in the environment are nitrogen cycle, oxygen cycle, carbon cycle, water cycle, etc.
- ii. Environmental balance is maintained through continuous operation of various cycles.

14. Which factors affect the environment? How? – Exercise Question 6 - a

15. What will happen if number of consumers in environment goes on increasing gradually?

If number of consumers go on increasing, it will affect the population of producers and other consumers disturbing the balance of the ecosystem.

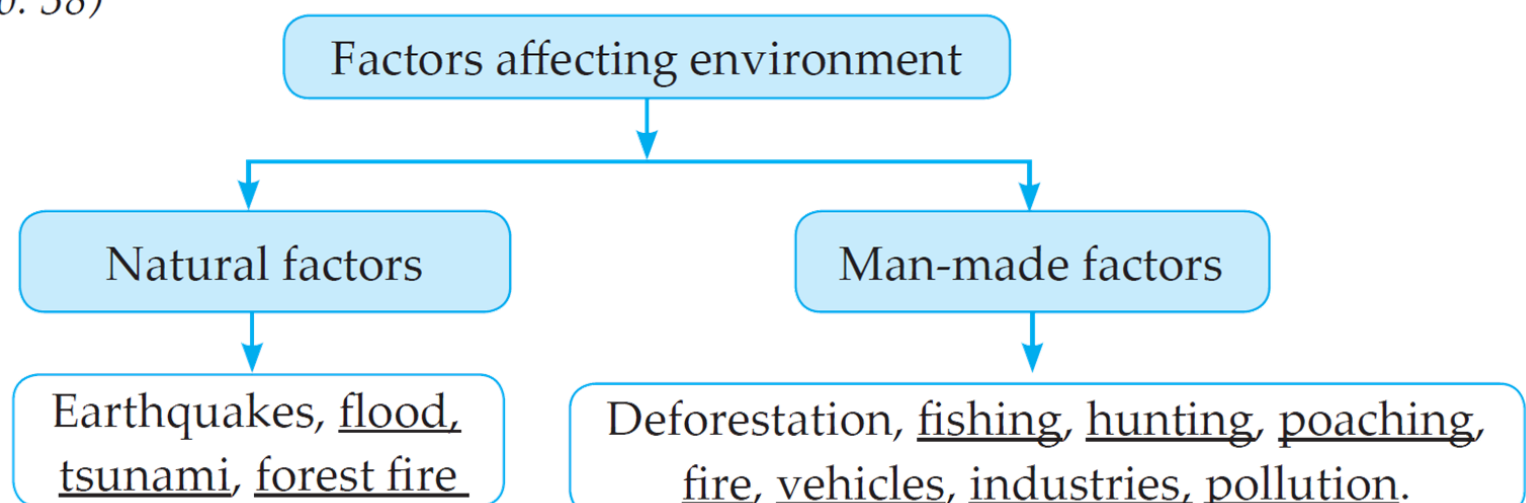
16. What will be the effect of industry established on riverbank on the river ecosystem?



- i. The river water will be affected by the chemicals released as effluents from the industry.
- ii. The animals and plants in the river may die.
- iii. The animals and human beings dependent on the river water also are seriously affected by the polluted water and it will make the soil around the polluted water infertile.

17. Complete the following flow chart.

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## 18. Which are the types of pollution?

**The types of pollution are:**

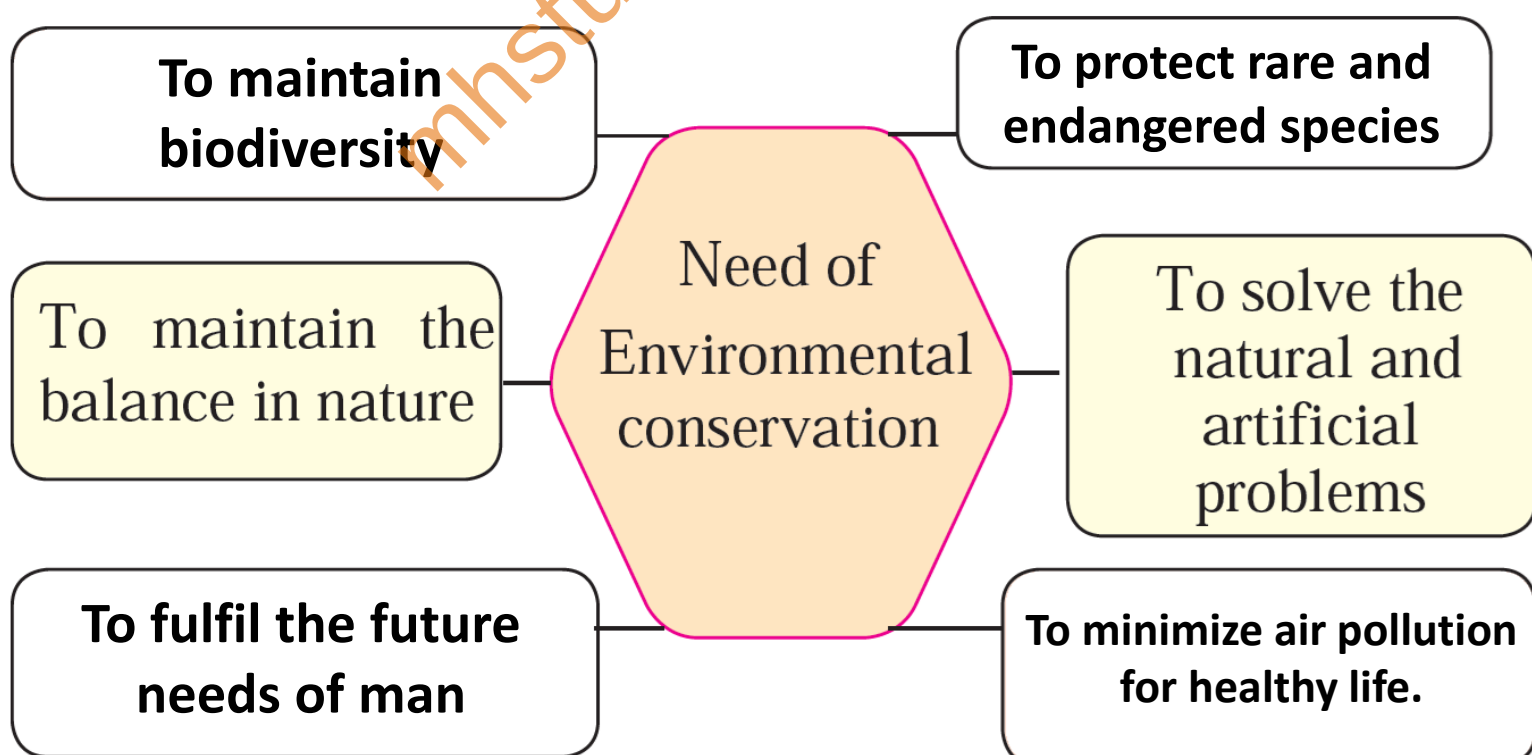
Air pollution, water pollution, soil pollution, radioactive pollution, thermal pollution, light pollution, noise pollution.

## 19. What do we mean by natural and artificial pollution?

- i. Unnecessary and unacceptable change in the surrounding environment due to natural events is called natural pollution. e.g., Forest fires, volcanoes, tsunami, etc.
- ii. Unnecessary and unacceptable change in the surrounding environment due to human activities is known as artificial pollution. e.g., industrialization, deforestation, unplanned urbanization, etc.

## 20. Why is it said that pollution control is important? – Exercise Question 4

## 21. Complete the Chart







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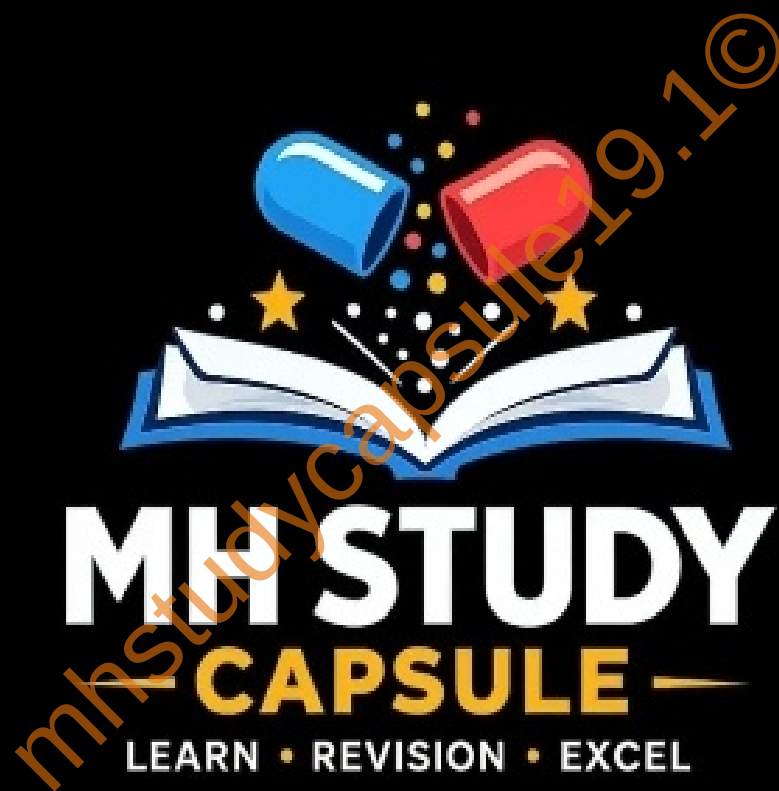
PYQs



Practice

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## 23. How do butterflies contribute to environmental balance?

**Butterflies contribute to environmental balance in several ways:**

- i. **Pollination**: Butterflies are excellent pollinators. As they move from flower to flower to drink nectar, pollen sticks to their bodies and gets transferred to other flowers, helping plants reproduce and grow more fruits and seeds.
- ii. **Biodiversity**: Butterflies are a part of the food chain. They are food for birds and other animals, helping maintain a balance in the ecosystem. When they thrive, they support the survival of other species as well.
- iii. **Indicator Species**: The presence and health of butterflies can indicate the health of an ecosystem. If butterflies are disappearing or declining in numbers, it might be a sign of environmental problems like pollution or habitat loss, alerting us to take action to protect the ecosystem.

### **REMEMBER THIS**

- ✓ **Forest (Conservation) Act, 1980**
- ✓ **Environmental (Protection) Act, 1986**
- ✓ **Sound Pollution (Control & Prevention) Rule, 2000.**
- ✓ **Biomedical Waste (Management & Handling) Rule, 1998.**
- ✓ **E-waste (Management & Handling) Rule, 2011.**



**Always remember**

**As per wildlife protection Act 1972**

**As per clause 49 A**, trading of rare animals has been completely banned.

**As per clause 49 B**, use of articles prepared from skin or organs of wild animals has been banned.

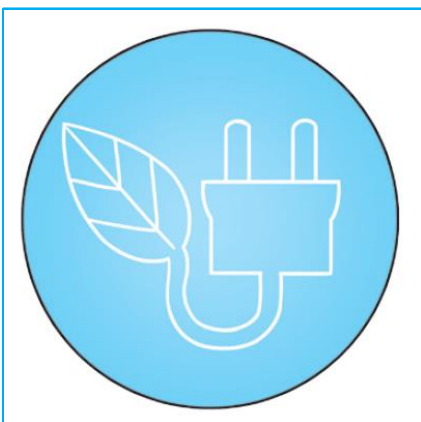
**As per clause 49 C**, disclosure of the stock of artefacts made from rare wild animals is compulsory.



**24. Find the meaning of these symbols in relation to environment conservation.**



This symbol tells us to 'Use Dustbin'. The garbage should not be thrown anywhere. It should be properly disposed.



This symbol tells us to 'Save Electricity'. If electricity is carefully used, we can conserve our natural resources.



This symbol tells us to 'Use Green Energy'. By riding on a bicycle, we save on fuel and use our own muscular energy. It is the best eco-friendly, non-polluting vehicle.



This symbol is giving the message "Save water". Sustainable use of water is necessary for our future.

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# Previous Year Questions

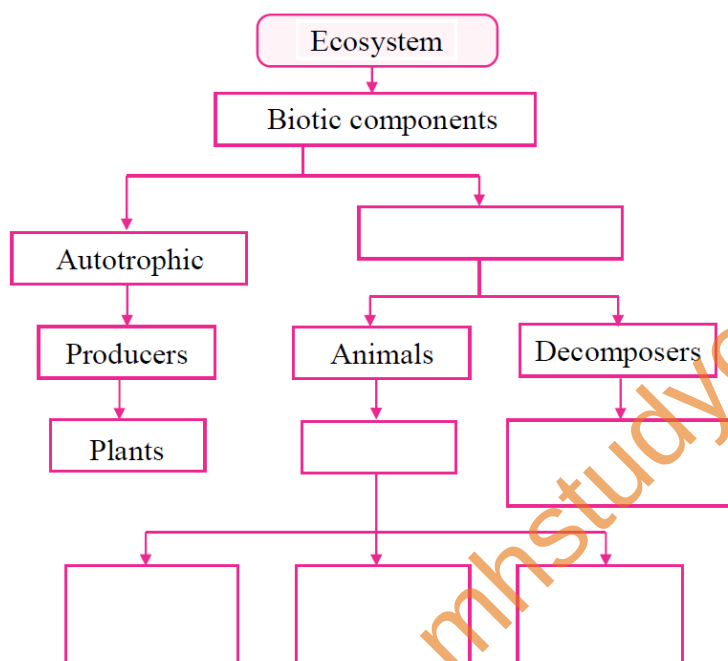
1. a. Identify the following symbols and state their significance:



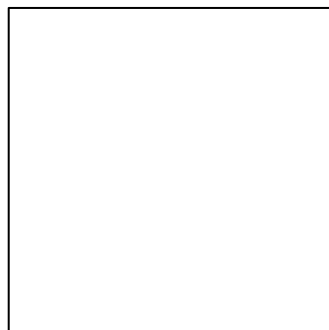
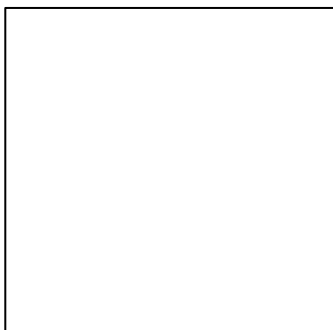
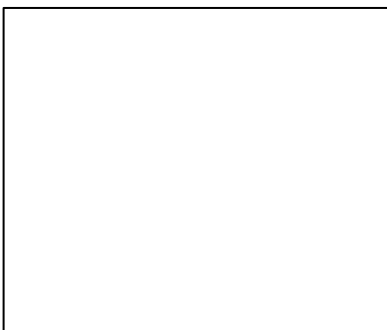
Intext Question 24

b. How can biodiversity be conserved? – Exercise Question 6 - d

2. Complete the flow chart: - Intext Question 8

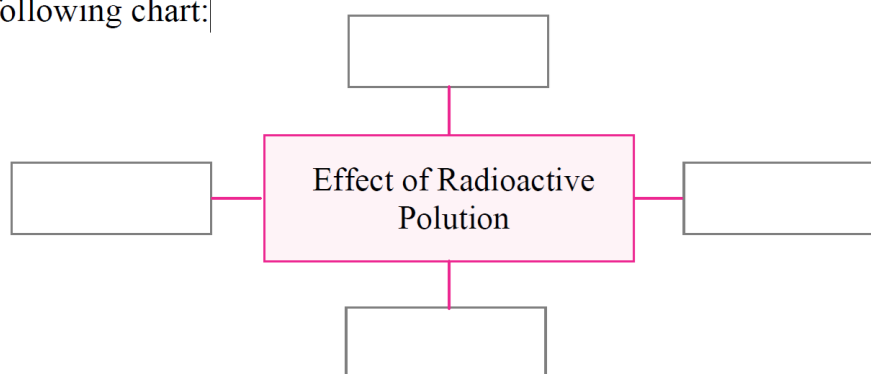


3. What are the meaning of the following symbols? Write your role accordingly. – Exercise Question 7



#### 4. Complete the following chart:

Following chart:



- i. Cancerous ulceration occurs due to higher radiations of X-rays.
- ii. Tissues in the body are destroyed.
- iii. Genetic changes occur.
- iv. Vision is adversely affected.

#### 5. Attempts at various levels are performed for conserving environment.

Which role would you like to perform. Give two actions each:

- a. Prevention
- b. Control
- c. Production
- d. Awareness
- e. Conservation

I would like to play the role in spreading awareness about the conservation of the environment. By spreading awareness, we can change the mind set and approach of people towards the environment.

##### a. Prevention

- i. I will try to prevent possible harms to environment.
- ii. I will not use plastics to prevent pollution caused by it.

- i. I will use water and other natural resources in only limited required quantity.
- ii. I will stop harmful activities like throwing garbage on the road or in public places.

**c. Production**

- i. I will work for the revival of environmental factors.
- ii. I will try to develop projects for the conservation of special species.

**d. Awareness**

I will spread awareness through education, guidance, participation in various activities.

**e. Conservation**

- i. I will help in conserving the available resources.
- ii. I will protect plants and animals in my locality.

**6. Write a short note on Sacred groves. – Exercise Question 3 - d**

**7. Explain the three different levels of Biodiversity. – Exercise Que 6 - c**

**8. Define endangered species. Give two examples.**

Endangered species are organisms whose numbers have declined or whose habitats have shrunk to such an extent that they could become extinct in the near future if conservation measures are not implemented.

Examples include the lion-tailed monkey and the lesser florican.

## 9. What are the three endangered heritage places in India?

- i. The Western Ghat spread over the states of Gujarat, Maharashtra, Goa, Karnataka, Tamil Nadu and Kerala has been endangered due to mining industry and search for natural gas.
- ii. Habitats of Asiatic lion and wild bison of this region have been under threat.
- iii. Manas National Park of the Assam is under threat due to dams and indiscriminate use of water. Tiger and rhino of that region are under threat.
- iv. Sunderban National Park of West Bengal is reserved for tigers. However, the tiger population and overall local environment is seriously challenged by dams, deforestation, excessive fishing, trenches dug for same, etc.

## 10. What do we learn from the story of Jadav Molai Payeng? **Exercise**

### **Question 6 - e**

## 11. Explain the meanings of the following symbols:



### **Intext Question 24**

## 12. Write a short note on Environmental Conservation. **Exercise Question**

### **3 - a**

## 13. Write the factors affecting environment. **Exercise Question 6 - a**